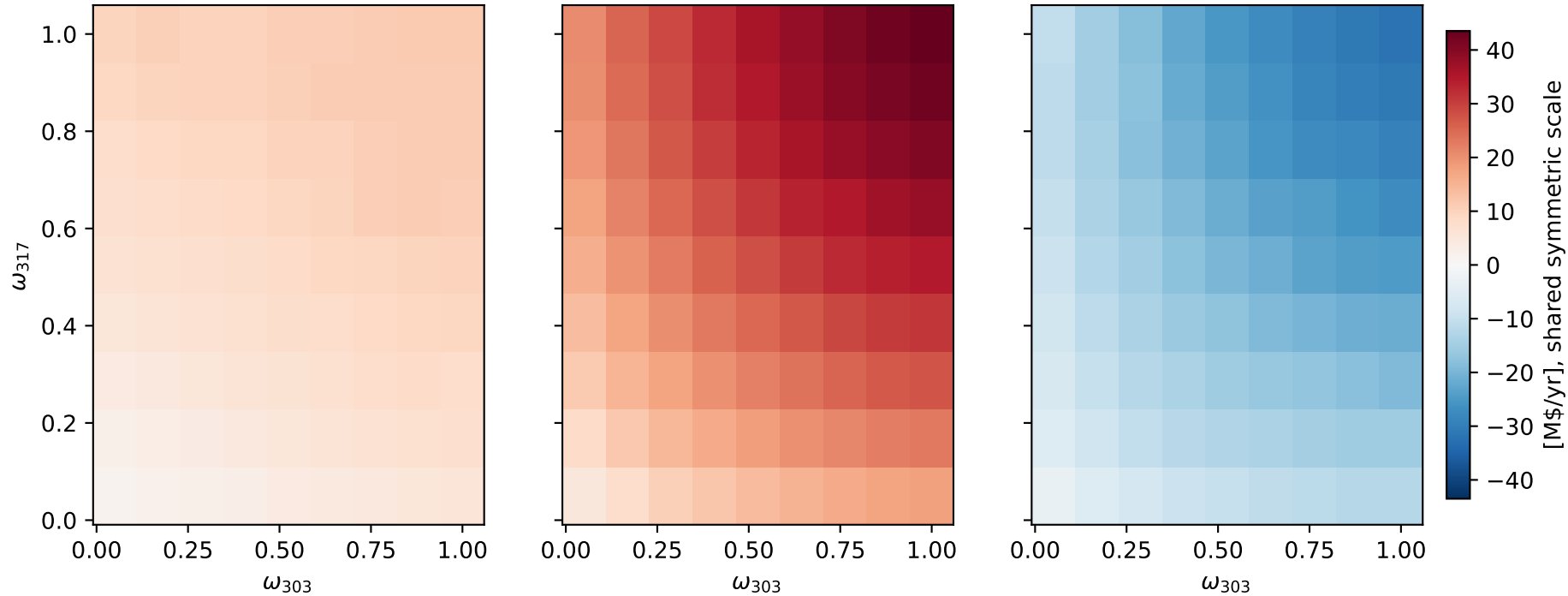


Cost decomposition, scenario A ($\rho_{H2} = 1.0$ \$/kg): the APPROX-column sign structure is raw minus synthetic bid cost
raw Δ cost = total - base K_{syn} (synthetic bid cost) APPROX = raw - K_{syn}



Identity check: $\max |\text{raw} - K_{syn} - \text{less-synthetic}| = 5.96\text{e-}08$ USD over 81 runs (holds to < 1 USD). Raw cost is positive in every run; negative APPROX-column values arise from the principled synthetic-bid subtraction (19b SB1) — report them only as APPROX-column statements with this decomposition displayed.